



Department of Information Technology

Academic Year 2024 – 2025(Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CS3591 & Computer Networks

Name of the Faculty member (s): Dr. K. Palraj, ASP / IT

Innovative Practice Description

- **Unit / Topic:** Unit 3 / Subnetting
- **Course Outcome:** The student will be able to know the idea of the subnetting.
- **Topic Learning Outcome:** TL024.
- **Activity Chosen:** Flipped Class Room
- **Justification:**
 - To determine the students' level of comprehension regarding the subnetting, a video describing the concept of subnetting was posted to the student group. The question was asked in the majority of university questions, and it is one that is important.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The material for the activity were given to the students one week before through the mail and canvas. The students were asked to watch the video individually at their home and take the notes according to their understanding level.
 - On the day of activity ask the students to form a group. Each student discussed the topic subnetting with example with their group.
 - Mr.C.Gowtham and Mr.M. Joel Paulraj team explained the concept of subnetting in Figure 1.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	3	2	2	3	1	2	2	1	2	1	1	3
(1 – Low 2 – Moderate 3 – High)													

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	2	2	3	1	2	2	1	2	1	1	3
Justification for correlation	The students will be able to employ basic mathematical skills in applications related to the network layer.	The students could be able to formulate and analyze the various protocols for network layer mapped with real-time applications	The students could be able to design the various switching for engineering problems and real time problems	The students could be able to interpret and analyze the real-world application using Internet protocols	The students could be able to use internet addressing for data transfer	The students will complete the assessment in an ethical way	The students will work as part of a team and individually to illustrate their competencies in the network layer	The students will illustrate their abilities in communication and problem-solving as they engage with the network layer.	The students will be able to recognize the tasks necessary for developing the IPv6 protocol.	The student will identify technological developments from the past and enhance protocols within the network layer.	The students would be able to provide a solution for the address resolution protocol using their computing skills	The students could be able to provide the IT solution with network addressing with network layer	The students could be able to develop application with network layer

- **Images / Screenshot of the practice:**



Figure 1: Discussion on the topic Subnetting C. Gowtham and M. Joel Pulraj Team

- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***

- The students are actively participated in the activity.
 - The student can able to explore their knowledge on explaining the topic subnetting and its usage in LAN Setup.
 - This activity helps the students to gain knowledge on the process of dividing a network into smaller subnetworks, or subnets, to allow devices within each subnet to communicate directly.

- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The students understand well about subnetting in the network.
 - It worked well for the students to interact with the teacher and other students during the lesson.
 - Students' fear decreased when they talked to their peers about the topic.
 - It helps to developing their self-learning ability.
 - All the students demonstrate a thorough understanding of subnetting.

- ❖ ***Challenges faced in implementation:***

- Some students are hesitated to talk in front of all the students.
 - The activity was planned for 10 Minutes, but it takes 15 minutes because the students hesitate to come dais.

References:

- <https://www.ritrjpm.ac.in/departments/rit-information-technology/it-innovative-practices.php>
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/3_CS3352_VA_FPR.pdf

Signature of Faculty Member

HOD